Technical Supplement: How Much History Is Enough? Certifying Decision-Relevant Memory in Offline Reinforcement Learning

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Notation

At stage h, X_h is the full observable history and $\phi_{m,h}(X_h)$ is its m-observation suffix. The suffix cell is $C_{m,h}(z) = \{x : \phi_{m,h}(x) = z\}$. We use $A_h^*(x, a) = V_h^*(x) - Q_h^*(x, a)$,

$$\eta_{m,h}(z) = \min_{p \in \Delta(\mathcal{A}_h)} \max_{x \in C_{m,h}(z)} \mathbb{E}_{a \sim p} A_h^*(x, a),$$

$\eta_{m,h} = \max_z \eta_{m,h}(z)$, and $G_m = \sum_h \eta_{m,h}$.

Population Proofs

Lemma 1 (Monotonicity)**.** *For every stage, $\eta_{m+1,h} \le \eta_{m,h}$, and therefore $G_{m+1} \le G_m$.*

Proof. Every $(m+1)$-suffix cell C' lies inside one m-suffix cell C. Let p_C minimize the objective for C. Then

$$\begin{aligned} \eta(C') &\le \max_{x \in C'} \mathbb{E}_{a \sim p_C} A_h^*(x, a) \\ &\le \max_{x \in C} \mathbb{E}_{a \sim p_C} A_h^*(x, a) = \eta(C). \end{aligned}$$

Taking the maximum over refined cells proves the stagewise claim; summing proves the cumulative claim. □

Theorem 1 (Exact sufficiency)**.** *For a fixed m, $G_m = 0$ if and only if there exists an m-suffix policy that is optimal from every full context at every stage.*

Proof. All entries of A_h^* are nonnegative. Thus a cell has zero minimax value if and only if some action distribution $p_{h,z}$ has zero expected advantage loss at every context in the cell. Use these distributions to form a suffix policy π. At stage H, $V_H^\pi(x) = \mathbb{E}_{a \sim p_{H,z}} Q_H^*(x, a) = V_H^*(x)$. If the equality holds at stage $h+1$, then

$$\begin{aligned} V_h^\pi(x) &= \mathbb{E}_{a \sim p_{h,z}}[r_h(x, a) + P_h V_{h+1}^\pi(x, a)] \\ &= \mathbb{E}_{a \sim p_{h,z}} Q_h^*(x, a) = V_h^*(x). \end{aligned}$$

This proves sufficiency by backward induction.

Conversely, suppose π is uniformly optimal. Backward induction from the terminal stage shows that its continuation after every action is also valued by V_{h+1}^*, hence

$$V_h^*(x) = V_h^\pi(x) = \mathbb{E}_{a \sim \pi_h(z)} Q_h^*(x, a).$$

The expected optimal advantage loss is zero in every context of every cell, so $G_m = 0$. □

Theorem 2 (Value-loss sandwich)**.** *Let $\mathcal{E}_m = \inf_{\pi \in \Pi_m} \max_{h,x}[V_h^*(x) - V_h^\pi(x)]$. Then*

$$\max_h \eta_{m,h} \le \mathcal{E}_m \le \sum_h \eta_{m,h}.$$

Proof. For the upper bound, let π^m use a minimax action distribution in each cell and define $D_h(x) = V_h^*(x) - V_h^{\pi^m}(x)$. Since $Q_h^*(x, a) = r_h(x, a) + P_h V_{h+1}^*(x, a)$,

$$\begin{aligned} D_h(x) &= \mathbb{E}_{a \sim \pi_h^m(z)} A_h^*(x, a) + \mathbb{E}_{a, X'} D_{h+1}(X') \\ &\le \eta_{m,h} + \|D_{h+1}\|_\infty. \end{aligned}$$

With $D_{H+1} = 0$, backward induction yields $D_h(x) \le \sum_{t=h}^H \eta_{m,t}$.

For the lower bound, fix any $\pi \in \Pi_m$. Since $V_{h+1}^\pi \le V_{h+1}^*$,

$$V_h^\pi(x) \le \mathbb{E}_{a \sim \pi_h(z)} Q_h^*(x, a),$$

and therefore

$$V_h^*(x) - V_h^\pi(x) \ge \mathbb{E}_{a \sim \pi_h(z)} A_h^*(x, a).$$

For each stage, maximize over contexts within cells and then over cells. Any shared action distribution incurs at least $\eta_{m,h}$, so the uniform loss is at least $\max_h \eta_{m,h}$. Taking the infimum over π completes the proof. □

Observation Prediction versus Decision Sufficiency

Fix a behavior policy μ and define

$$\begin{aligned} R_m^{\text{obs}} = &\sum_h H_\mu(O_{h+1} \mid \phi_{m,h}(X_h), A_h) \\ &- \sum_h H_\mu(O_{h+1} \mid X_h, A_h). \end{aligned}$$

In the delayed-cue construction, all observations after the initial cue are constant, so $R_m^{\text{obs}} = 0$ for every m, whereas the final action incurs minimax advantage loss $\gamma/2$ until the cue enters the suffix. In the nuisance construction, the initial bit reappears as the terminal observation but action zero is optimal after either bit. Hence $G_m = 0$ for every m, while $R_m^{\text{obs}} = 1$ bit for all $m < H$ under a balanced cue.

This incomparability concerns observation-only prediction. The exact controlled Markov condition is stronger. Suppose that for each h, m, z, a, the conditional joint law of $(R_h, \phi_{m,h+1}(X_{h+1}))$ is identical for all $x \in C_{m,h}(z)$. Backward induction then shows that $Q^*_h(x, a)$ is constant within every suffix cell for each action. A common maximizing action therefore exists, and $G_m = 0$. Thus full reward–transition predictive sufficiency upper-bounds decision-relevant memory; the potentially two-sided failures arise for observation-prediction diagnostics.

Offline Certificate Proofs

Assume simultaneous intervals $L_h \le Q^*_h \le U_h$ and define

$$\overline{A}_h(x, a) = \max\{0, \max_{b \neq a}[U_h(x, b) - L_h(x, a)]\}.$$

The self-comparison is omitted because it is exactly zero.

Lemma 2 (Upper advantage). *On the interval event, $A^*_h(x, a) \le \overline{A}_h(x, a)$ for all h, x, a.*

Proof.

$$\begin{aligned} A^*_h(x, a) &= \max\{0, \max_{b \neq a}[Q^*_h(x, b) - Q^*_h(x, a)]\} \\ &\le \overline{A}_h(x, a). \end{aligned}$$

□

Theorem 3 (Simultaneous value certificate). *Let $\overline{\pi}^m$ be the suffix policy obtained by replacing A^*_h with $\overline{A}_h$ in each cell-wise minimax problem. Then, on the interval event, simultaneously for all m, h, x,*

$$V^*_h(x) - V^{\overline{\pi}^m}_h(x) \le \sum_{t=h}^{H} \overline{\eta}_{m,t}.$$

Proof. The proof is the upper-bound recursion above, with

$$\mathbb{E}_{a \sim \overline{\pi}^m_h(z)} A^*_h(x, a) \le \mathbb{E}_{a \sim \overline{\pi}^m_h(z)} \overline{A}_h(x, a) \le \overline{\eta}_{m,h}.$$

The interval event is simultaneous across states, actions, stages, and memory lengths, so a data-dependent choice among the candidate memories remains covered. □

Proposition 1 (Interval tightness). *If every one-sided interval error is at most w, then*

$$A^*_h \le \overline{A}_h \le A^*_h + 2w, \quad \eta_{m,h} \le \overline{\eta}_{m,h} \le \eta_{m,h} + 2w,$$

and $G_m \le \overline{G}_m \le G_m + 2Hw$.

Proof. For every competitor $b \neq a$,

$$U_h(x, b) - L_h(x, a) \le Q^*_h(x, b) - Q^*_h(x, a) + 2w.$$

Taking the positive maximum gives the pointwise cost inequality. The lower LP inequality follows from pointwise domination. For the upper LP inequality, evaluate the robust objective at an exact minimizer; adding at most $2w$ to each action cost adds at most $2w$ to every mixture cost. Sum over stages. □

Corollary 1 (Adaptive localization and exact recovery). *Let $m^*_\tau = \min\{m : G_m \le \tau\}$, with the full-history candidate included. For every $\epsilon \ge 2Hw$, DRMS obeys*

$$m^*_\epsilon \le \widehat{m}_\epsilon \le m^*_{\epsilon - 2Hw}.$$

If $m_{\mathrm{dec}} = \min\{m : G_m = 0\} > 1$, let $\Gamma = \min_{m < m_{\mathrm{dec}}} G_m$. If $2Hw < \Gamma$, then every threshold $\epsilon \in [2Hw, \Gamma)$ makes DRMS select m_{dec} on the interval event.

Proof. Certification of $\widehat{m}_\epsilon$ implies $G_{\widehat{m}_\epsilon} \le \overline{G}_{\widehat{m}_\epsilon} \le \epsilon$, hence $m^*_\epsilon \le \widehat{m}_\epsilon$. Conversely,

$$\overline{G}_{m^*_{\epsilon - 2Hw}} \le G_{m^*_{\epsilon - 2Hw}} + 2Hw \le \epsilon,$$

so the shortest certified memory cannot exceed $m^*_{\epsilon - 2Hw}$. The full-history candidate ensures that the set is nonempty. For exact recovery, every shorter memory has $\overline{G}_m \ge G_m \ge \Gamma > \epsilon$, whereas $\overline{G}_{m_{\mathrm{dec}}} \le 2Hw \le \epsilon$. □

Deployment-Weighted Certificate Proofs

For a reference distribution ν_h over full histories, define

$$\ell^\nu_{m,h}(z, p) = \sum_{x \in C_{m,h}(z)} \nu_h(x) \mathbb{E}_{a \sim p} A^*_h(x, a),$$

$$g^\nu_{m,h} = \sum_z \min_{p \in \Delta(\mathcal{A}_h)} \ell^\nu_{m,h}(z, p), \qquad G^\nu_m = \sum_h g^\nu_{m,h}.$$

Because each cell objective is linear in p, a deterministic action is optimal except at ties.

Lemma 3 (Weighted monotonicity). *For every reference sequence ν, $g^\nu_{m+1,h} \le g^\nu_{m,h}$ and $G^\nu_{m+1} \le G^\nu_m$.*

Proof. An $(m+1)$-suffix partition refines the m-suffix partition. Any action rule feasible for the coarser partition remains feasible after assigning that same action to every refined subcell. Minimizing over the larger refined policy class cannot increase the reference-weighted objective. Sum over stages. □

Theorem 4 (Population deployment certificate). *Let $\pi^{m,\nu}$ attain the weighted cell objectives above, and let $d^{\pi^{m,\nu}}_h$ be its state occupancy from initial distribution ρ. If*

$$d^{\pi^{m,\nu}}_h(x) \le C_h \nu_h(x) \quad \text{for every } h, x,$$

then

$$V^*_1(\rho) - V^{\pi^{m,\nu}}_1(\rho) \le \sum_{h=1}^{H} C_h g^\nu_{m,h}.$$

Proof. The finite-horizon performance-difference identity, written with the optimal advantage loss, is

$$V^*_1(\rho) - V^\pi_1(\rho) = \sum_{h=1}^{H} \mathbb{E}_{X_h \sim d^\pi_h,\, A_h \sim \pi_h} A^*_h(X_h, A_h).$$

All summands are nonnegative. The density-ratio assumption therefore gives

$$\mathbb{E}_{d^\pi_h, \pi_h} A^*_h \le C_h \mathbb{E}_{\nu_h, \pi_h} A^*_h.$$

For $\pi = \pi^{m,\nu}$, the right-hand expectation is exactly $g^\nu_{m,h}$. Summing proves the claim. □

Theorem 5 (Empirical occupancy and robust-cost correction)**.** *Suppose the simultaneous Q^*-interval event holds and $0 \le \overline{A}_h(x,a) \le B_h$. Let $\widehat{\nu}_h$ satisfy $\|\widehat{\nu}_h - \nu_h\|_1 \le \xi_h$, and let $\widehat{\pi}_m$ minimize the $\widehat{\nu}$-weighted robust costs. If its deployment occupancy obeys $d_h^{\widehat{\pi}_m}(x) \le C_h \nu_h(x)$, then*

$$V_1^*(\rho) - V_1^{\widehat{\pi}_m}(\rho) \le \sum_{h=1}^{H} C_h \left(\widehat{\overline{g}}_{m,h} + \frac{B_h \xi_h}{2}\right).$$

For the clipped tabular intervals in this paper, one may take $B_h = H - h + 1$.

Proof. Fix a stage and define $f_h(x) = \mathbb{E}_{a\sim\widehat{\pi}_{m,h}(\phi_m(x))}\overline{A}_h(x,a)$. Then $f_h \in [0, B_h]$. Total-variation duality yields

$$\mathbb{E}_{\nu_h} f_h \le \mathbb{E}_{\widehat{\nu}_h} f_h + \frac{B_h}{2}\|\widehat{\nu}_h - \nu_h\|_1 \le \widehat{\overline{g}}_{m,h} + \frac{B_h \xi_h}{2}.$$

The first inequality also holds for the data-selected f_h, because the L_1 event controls every bounded function simultaneously. Combine $A_h^* \le \overline{A}_h$, the deployment-to-reference density-ratio bound, and the performance-difference identity, then sum over stages. □

For n independent episodes, the empirical stage-h state frequency $\widehat{\nu}_h$ satisfies the simultaneous Weissman-style radius

$$\xi_h = \min\left\{2, \sqrt{\frac{2\{|\mathcal{X}_h|\log 2 + \log(H/\delta_{\mathrm{occ}})\}}{n}}\right\}.$$

The constants C_h must be known from the deployment design or replaced by valid upper bounds. In the rare-cue weighted experiment, actions do not affect state occupancy, so $C_h = 1$ exactly. When no finite or credible bound is available, the uniform DRMS certificate should be used.

Tabular Confidence Intervals

Let $M = \sum_h |\mathcal{X}_h||\mathcal{A}_h|$. For visit count $N_h(x,a)$, use reward and transition radii

$$\beta_h^r = \sqrt{\frac{\log(4M/\delta)}{2\max\{1, N_h\}}},$$

with radius one if $N_h = 0$, and

$$\beta_h^P = \min\left\{2, \sqrt{\frac{2(|\mathcal{X}_{h+1}|\log 2 + \log(2M/\delta))}{\max\{1, N_h\}}}\right\}.$$

Hoeffding's inequality, the L_1 deviation inequality of Weissman et al. (2003), and a union bound give simultaneous model coverage.

Initialize $\underline{V}_{H+1} = \overline{V}_{H+1} = 0$ and recurse

$$\underline{Q}_h = \mathrm{clip}\left(\widehat{r}_h - \beta_h^r + \widehat{P}_h \underline{V}_{h+1} - \tfrac{1}{2}\beta_h^P \,\mathrm{span}(\underline{V}_{h+1})\right),$$
$$\overline{Q}_h = \mathrm{clip}\left(\widehat{r}_h + \beta_h^r + \widehat{P}_h \overline{V}_{h+1} + \tfrac{1}{2}\beta_h^P \,\mathrm{span}(\overline{V}_{h+1})\right),$$

where clipping is to $[0, H-h+1]$ and each value bound is the actionwise maximum of its Q-bound.

Proposition 2 (Validity)**.** *On the simultaneous model event, $\underline{Q}_h \le Q_h^* \le \overline{Q}_h$ at every state-action-stage triple.*

Proof. Proceed backward. If $\underline{V}_{h+1} \le V_{h+1}^* \le \overline{V}_{h+1}$, then

$$\begin{aligned} Q_h^* &= r_h + P_h V_{h+1}^* \\ &\ge \widehat{r}_h - \beta_h^r + P_h \underline{V}_{h+1} \\ &\ge \widehat{r}_h - \beta_h^r + \widehat{P}_h \underline{V}_{h+1} - \tfrac{1}{2}\beta_h^P \,\mathrm{span}(\underline{V}_{h+1}). \end{aligned}$$

The upper inequality is symmetric. Maximization preserves value bounds, and clipping preserves the bounds because remaining returns lie in $[0, H-h+1]$. □

Matching Rates in the Two-Point Family

Let $B \sim \mathrm{Bernoulli}(p)$ be an observed stage-one cue and let the stage-two current observation be constant. The behavior policy chooses the two final actions uniformly. Write

$$r_+ = \tfrac{1}{2} + \tfrac{\gamma}{2}, \qquad r_- = \tfrac{1}{2} - \tfrac{\gamma}{2}.$$

In model $\mathcal{M}_0$, action zero has Bernoulli mean r_+ after either cue and action one has mean r_-. In model $\mathcal{M}_1$, these means are swapped only after $B = 1$. Therefore $m_{\mathrm{dec}}(\mathcal{M}_0) = 1$ and $m_{\mathrm{dec}}(\mathcal{M}_1) = 2$.

Theorem 6 (Lower bound)**.** *For $0 < \gamma \le 1/2$ and $0 < \delta < 1/4$, any estimator with error at most δ in both models requires*

$$n \ge \frac{\log(1/(4\delta))}{6p\gamma^2}.$$

Proof. The models differ only after the rare cue. Under the uniform final action, the one-episode divergence is

$$D_{\mathrm{KL}}(P_0\|P_1) = \frac{p}{2}\left[\mathrm{kl}(r_+\|r_-) + \mathrm{kl}(r_-\|r_+)\right].$$

For Bernoulli variables, $\mathrm{kl}(u\|v) \le (u-v)^2/[v(1-v)]$. Since $r_- r_+ = 1/4 - \gamma^2/4 \ge 3/16$, each KL term is at most $(16/3)\gamma^2$. Thus $D_{\mathrm{KL}}(P_0\|P_1) \le (16/3)p\gamma^2 < 6p\gamma^2$.

The Bretagnolle–Huber inequality gives, for any test,

$$\mathbb{P}_0(\text{error}) + \mathbb{P}_1(\text{error}) \ge \tfrac{1}{2}\exp[-D_{\mathrm{KL}}(P_0^n\|P_1^n)].$$

If both errors are at most δ, then $2\delta \ge \frac{1}{2}e^{-6np\gamma^2}$. Rearrangement proves the result. □

Corollary 2 (Tabular DRMS upper bound)**.** *Assume $p \le 1/2$, and let M be the total number of full-history state–action pairs. Run the tabular interval construction with confidence parameter $\delta/2$, and run DRMS with tolerance $\gamma/4$. If*

$$n \ge \frac{32\log(8M/\delta)}{p\gamma^2},$$

then DRMS returns the correct memory in both models with probability at least $1-\delta$.

Proof. Every final state–action count has binomial mean at least $np/2$. A multiplicative Chernoff bound and a union bound over the four final pairs give

$$\mathbb{P}\left(\min_{x,a} N_2(x,a) < np/4\right) \le 4e^{-np/16}.$$

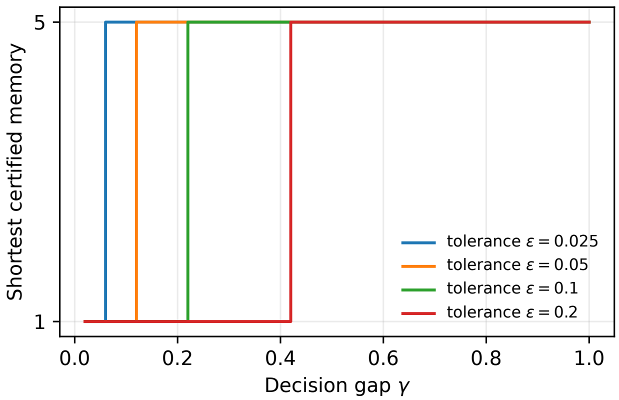

Figure 1: Population tolerance phase transition at horizon five. An insufficient suffix has ambiguity $\gamma/2$, so memory one is certified exactly when $\gamma \leq 2\epsilon$.

The stated sample size, together with $\gamma \leq 1/2$, makes this at most $\delta/2$. On the simultaneous reward-concentration event, which has probability at least $1 - \delta/2$, every final reward radius obeys

$$\beta_2^r(x, a) \leq \sqrt{\frac{2\log(8M/\delta)}{np}} \leq \frac{\gamma}{4}.$$

Because the empirical mean can deviate from the truth by at most the same radius, $L_2(x, a_*) \geq Q_2^*(x, a_*) - 2\beta_2^r$ and $U_2(x, a) \leq Q_2^*(x, a) + 2\beta_2^r$. Hence the true best action in each final context has robust upper advantage zero. In $\mathcal{M}_0$, action zero is common to both contexts, so $\overline{G}_1 = 0$. In $\mathcal{M}_1$, $\overline{G}_1 \geq G_1 = \gamma/2 > \gamma/4$, while each full-memory cell has $\overline{G}_2 = 0$. DRMS therefore selects one and two, respectively. □

Additional Experimental Details

The random generator uses binary observations and actions. Initial observations and action-conditional next observations are sampled from Dirichlet$(1, 1)$, and each state-action mean reward is sampled from Beta$(2, 2)$. Full action–observation histories remain distinct states, so short suffixes induce random aliasing. We use horizons three and four, 100 seeds per horizon, and synthetic Q^* interval half-width $w = 0.05$. Across all 700 memory configurations, there are no failures of uniform monotonicity, either side of the population sandwich, the robust certificate, the $2Hw$ inflation bound, the weighted population certificate, or the weighted robust certificate. The exact one-sided 95% binomial upper bound on the violation probability after observing zero failures is $1 - 0.05^{1/700} = 0.427\%$. Among configurations with nonzero realized loss, the median relative slack of the uniform population upper bound is 24.6%, with 90th percentile 65.5%.

Computational Reproducibility

All stochastic scripts construct seeds deterministically from the displayed grid indices and expose the number of repetitions as a command-line argument. For the tabular trajectory datasets, episodes are simulated in vectorized stage batches and reduced to state–action counts, Bernoulli reward sums, and transition counts. This retains exact episode sampling while keeping the largest sweep lightweight. The reported artifacts were regenerated on Debian GNU/Linux 13 using five AMD EPYC 9V74 vCPUs and 5.9 GiB RAM. The software stack was Python 3.13.5, NumPy 2.3.5, SciPy 1.17.0, pandas 2.2.3, Matplotlib 3.10.8, and pytest 9.0.2; continuous integration independently runs Python 3.11. No GPU is used. The repository includes raw CSVs, generated figures, 16 theorem-driven tests, and a single command that rebuilds all results and both manuscripts.

The experimental metrics are certificate validity, simultaneous interval coverage, false short-memory certification, correct-memory certification, selected memory, realized value loss, and certificate slack. These metrics were chosen to test the finite-sample statements directly rather than to claim benchmark dominance. Recovery and certification rates are reported over independent seeds; for the zero-violation randomized test we additionally report an exact one-sided binomial bound. Significance tests for pairwise algorithm improvements are not used because the paper makes no such performance-comparison claim.

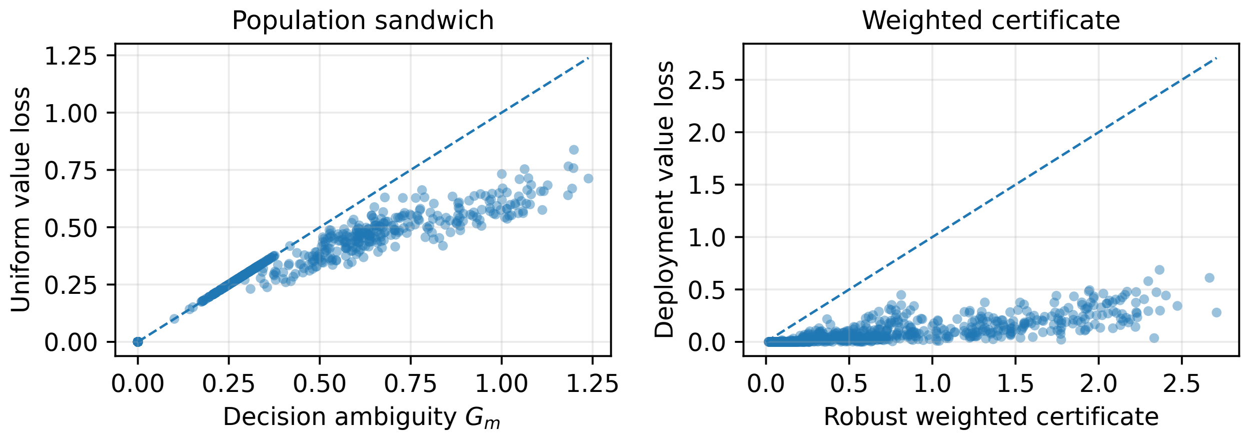

Figure 2: **Random history-MDP stress test.** Left: the realized uniform value loss of the population minimax policy lies below G_m. Right: the realized deployment loss of the robust weighted policy lies below its weighted certificate. Dashed lines denote equality. The suite contains 700 memory configurations from 200 random history MDPs.

Experiment	Complete grid and repetitions
Population separation	Tasks: delayed cue and predictive nuisance; horizons $\{2, 3, 4, 5, 6, 8\}$; all candidate memories; balanced cue; gap one.
Soft relevance	Horizon five; 50 equally spaced gaps in $[0.02, 1]$; $\epsilon \in \{0.025, 0.05, 0.1, 0.2\}$.
Uniform calibration	Horizon four; $p \in \{0.05, 0.1, 0.2, 0.5\}$; $n \in \{50, 100, 200, 500, 1000, 2000, 5000, 10000\}$; gap 0.5; $\epsilon = 0.1$; $\delta = 0.05$; 50 seeds per point (1,600 datasets).
Coverage–gap scaling	Horizon two; $p \in \{0.05, 0.1, 0.2, 0.5\}$; $\gamma \in \{0.2, 0.3, 0.5\}$; eight sample sizes from 100 to 20,000; $\epsilon = \gamma/4$; $\delta = 0.05$; 30 seeds (2,880 datasets).
Weighted certificate	Gap 0.4, $\epsilon = 0.1$. Population: 20 logarithmically spaced values of p from 0.01 to 0.5. Finite sample: $p \in \{0.05, 0.1, 0.2, 0.5\}$, $n \in \{1000, 2500, 5000, 10000, 25000, 50000\}$, 15 seeds; $\delta = 0.05$ split equally between Q and occupancy events (360 datasets).
Random stress	Horizons $\{3, 4\}$; 100 seeds per horizon; every memory; interval half-width 0.05 (200 MDPs and 700 configurations).

Table 1: All grids used in the reported experiments. No result-dependent hyperparameter search is performed.

References

Weissman, T.; Ordentlich, E.; Seroussi, G.; Verdú, S.; and Weinberger, M. J. 2003. Inequalities for the L1 Deviation of the Empirical Distribution. Technical Report HPL-2003-97R1, Hewlett-Packard Laboratories.